Mufakir Qamar Ansari

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LinkedIn | *Github* | *Google Scholar* | *Website* | *ORCiD*

Summary

Data Scientist with 9+ years of combined research and industry experience in predictive modeling, statistical analysis, and applied machine learning. Builds end-to-end ML and ETL pipelines and applies generative AI, NLP, and deep learning to complex, high-dimensional datasets on distributed computing platforms. Translates analytical findings into actionable insights for cross-functional stakeholders and executive leadership across diverse domains.

Skills & Tools

Programming: Python, SQL, PySpark, R, C++, CUDA
Data Science & ML: Predictive Modeling, Generative AI, Deep Learning, NLP, Time Series Forecasting, Computer Vision, A/B Testing, Causal Inference, Statistical Modeling
Frameworks & Libraries: PyTorch, TensorFlow, Scikit-learn, LangChain, Hugging Face
Platforms & MLOps: AWS (SageMaker, S3), Azure Databricks, Apache Airflow, Docker, MLflow, Git, SLURM
Databases & Visualization: PostgreSQL, MySQL, MongoDB, ChromaDB, Neo4j, Tableau

Research Experience

Wright State University, *Dayton , Ohio* *Jan 2026 – Present*
Research Assistant

- Architected a 19-stage HPC ETL pipeline that processed 356 RNA-Seq samples and 15M+ read pairs by orchestrating SLURM-based parallel execution on the OSC supercomputer, with full reproducibility via AWS S3 and Azure Databricks.
- Engineered a generative AI benchmarking pipeline for HDLSS synthetic tabular data by training 7 deep generative models (CTGAN, TabDDPM, Flow Matching, TVAE) class-conditionally and validating output fidelity through Wasserstein distance, KS likeness, and Train-Synthetic-Test-Real (TSTR) classification across 7 classifiers.
- Developed DDFF, a distributional feature selection framework that achieved 99.7% dimensionality reduction on high-dimensional datasets by quantifying class-conditional PMF divergence in $O(pn)$ time; results accepted at IEEE ICMLA 2026.

The University of Toledo, *Toledo, Ohio* *Aug 2023 – Apr 2025*
Research Assistant, Transportation Systems Research Lab & High-Performance Computing Lab

- Delivered predictive crash-severity models for the Ohio Department of Transportation by processing 37,000+ geospatial records and engineering full-stack ensemble classifiers (Random Forest, XGBoost, LightGBM), producing a published state report (FHWA/OH-2025/19).
- Reduced deep learning training time by 50% (2× speedup) and accelerated GPU execution by 8.5% by optimizing distributed HPC workloads across SLURM-managed GPU clusters, while building a predictive scheduling model that cut the cluster's carbon footprint by 5%.

Professional Experience

Orcinus IT Solutions Private Limited, *Srinagar, India* *May 2018 - Dec 2022*
Technical Lead

- Reduced data pipeline latency by up to 35% for 7 SaaS enterprise clients by designing and deploying end-to-end ML solutions and scalable ETL workflows using Apache Airflow, PySpark, and Azure.
- Improved model reliability by 30% across production deployments by building a comprehensive MLOps framework enforcing version control, automated monitoring, and governance.

MyFajir IT Solutions Private Limited, *Srinagar, India*
Senior Engineer, then Consultant *Sep 2017 - Mar 2020*

- Processed 800+ daily transactions at 99.9% uptime by architecting production-grade RESTful APIs (Flask/FastAPI) and interactive analytics dashboards for cross-functional stakeholder decision support.
- Reduced client operational overhead by 20% by engineering a cloud-based ERP platform on Azure with SQL-driven relational database integration and agile sprint delivery via JIRA.

PanunKart.com, *Srinagar, India* *Aug 2013 to Sep 2017*
Business Operations Lead

- Drove 150% sales growth by designing and implementing a data strategy grounded in A/B testing and market segmentation analysis, directly improving data-driven decision-making across the organization.

Education

Master of Science in Computer Science & Engineering *Aug 2023 – Aug 2025*
Track: Artificial Intelligence *GPA: 3.91/4.00*
The University of Toledo, Ohio, USA
Bachelor of Technology in Electronics & Communication Engineering *Jul 2009 – Jul 2013*
National Institute of Technology, Srinagar, India

Projects

- **High-Performance Computing Pipeline for Ebola RNA-Seq Analysis** [GitHub]
  - Architected an end-to-end, 19-stage ETL pipeline on the OSC supercomputer via SLURM array parallelization, processing 356 samples and over 15 million read pairs from the 2014 West African Ebola outbreak.
  - Engineered robust dual-quantification workflows (HISAT2, Kallisto) and orchestration logic, ensuring seamless terabyte-scale data processing and generating cross-validated transcriptomic analytics.
- **Privacy-Preserving AI Agent & Knowledge Graph Engine** [GitHub]
  - Engineered a highly secure, cross-platform AI agent ecosystem (Tauri/Rust, FastAPI, Next.js) featuring robust AES-256-GCM encryption and local LLM inference, guaranteeing strict data sovereignty.
  - Architected a real-time Neo4j Knowledge Graph powered by custom NLP pipelines to continuously ingest, map, and analyze complex cross-platform user activity (browser engagement, desktop context) into deeply personalized insights.
- **Production LLM & RAG Implementation: Hybrid Optimization System**[GitHub]
  - Orchestrated an end-to-end LLM system featuring a RAG pipeline and ChromaDB vector database for complex financial data processing and entity extraction.
  - Evaluated and benchmarked multiple foundation models (Phi 2, Mistral 7B, Zephyr 7B) by optimizing prompt engineering, vector search latency, and production monitoring protocols.
  - Synthesized production-ready LLM inference patterns, achieving superior risk-adjusted returns and robust performance stability.

Research Papers & Academic Service

- **M. Ansari**, L. Popp, T. Ghosh, "Distributional Discriminative Feature Filtering: A Scalable Framework for High-Dimensional Low-Sample-Size Biomedical Data", *Accepted in IEEE ICMLA 2026.*
- **M. Q. Ansari** and M. Ansari, "Disentangling Architecture, Strategy, and Adaptation in Day-Ahead Electricity Price Forecasting: A Multi-Model Benchmark on Volatile US Markets," *Under Review, Computational Economics (Springer Nature).*
- **M. Q. Ansari** and M. Ansari., "DynaCut: Bounding Classical Memory in Quantum Circuit Knitting via Resource-Aware Tensor Network Reconstruction", *Under Review, Quantum Machine Intelligence.*
- **M. Q. Ansari**, "Dy-Part: A Dynamic, Noise-Aware Scheduler for Optimizing Hybrid Quantum-Classical Algorithms", *Under Review, Quantum Machine Intelligence.* Preprint available: DOI:10.21203/rs.3.rs-8041248/v1.
- **M. Q. Ansari**, et al., "From Text to Returns: Using Large Language Models for Mutual Fund Portfolio Optimization and Risk-Adjusted Allocation", Preprint available: DOI: arXiv:2512.05907v1.
- **M. Q. Ansari**, et al., "High-Sensitivity Detection of Invasive Ductal Carcinoma via Domain-Specific SimCLR Pre-Training," *Under Review, Signal, Image and Video Processing.* Preprint available: DOI:10.21203/rs.3.rs-8031909/v1.
- E. Chou, R. Rai, **M. Ansari**, S. Khanal, A. Davenport, "From Data to Action: Leveraging Machine Learning/Artificial Intelligence to Guide Proactive Pedestrian and Cyclist Safety Initiatives," *Ohio Department of Transportation Final Report FHWA/OH-2025/19*, August 2025. Available: SJN 136837.
- Hasan, T., Hossain, A., **Ansari, M. Q.**, & Syed, T. H. (2025). "Enhanced Intrusion Detection in IIoT Networks: A Lightweight Approach with Autoencoder-Based Feature Learning." *Proceedings of the 10th International Conference on Internet of Things, Big Data and Security (IoTBDS 2025)*, pp. 207-214.
- **M. Q. Ansari**, et al., "Racing to Idle: Energy Efficiency of Matrix Multiplication on Heterogeneous CPU and GPU Architectures," *American Journal of Computer Science and Technology (AJCST),* ISSN Online: 2640-012X; ISSN Print: 2640-0111, 2025. Preprint available: DOI:10.21203/rs.3.rs-7890483/v1.
- **M. Q. Ansari**, et al., "Accelerating Matrix Multiplication: A Performance Comparison Between Multi-Core CPU and GPU", Preprint available: DOI:10.48550/arXiv.2507.19723.
- Peer Reviewer: Engineering Applications of Artificial Intelligence (Elsevier, 42 papers), Journal of Pathology Informatics, ICMLA 2026, ICECCME 2026 and ACDSA 2027.

Certifications

Excel to MySQL: Analytic Techniques for Business	*Duke University*
Distributed Computing with Spark SQL	*UC Davis*
SQL for Data Science Capstone Project	*UC Davis*
Innovations in Investment Technology: Artificial Intelligence	*University of Michigan*
Managing Machine Learning Projects	*Duke University*
Managing Big Data with MySQL	*Duke University*
Increasing Real Estate Management Profits: Harnessing Data Analytics	*Duke University*
Data Visualization and Communication with Tableau	*Duke University*
Data Analytics for Lean Six Sigma	*University of Amsterdam*
Business Metrics for Data-Driven Companies	*Duke University*